



Latent Tuberculosis

Clinical Case Quiz

A 44-year-old male with no significant PMH is your next patient on GCN. His employer, a nursing home, requires an annual QuantiFERON Gold blood test for TB screening and requested this test to be ordered on a prior GCN encounter. He is calling you because he received a positive result, indicating 'TB infection likely,' and asks you what this means.

He indicates he has a history of BCG vaccine in the Philippines, where he was born, and cannot do the tuberculin (PPD) skin test. On Review of Systems, he denies any persistent cough, hemoptysis, night sweats, unexplained weight loss, anorexia, or recent international travel. He also denies any recent known TB exposure or any household contact with someone with TB. On chart biopsy, you note the patient had a negative QuantiFERON Gold in the prior calendar year. What is the next best step in management?

- Order repeat blood test using QuantiFERON Gold to verify the result.
- Order a PPD skin test using PNL PPD to check for active TB.
- Provide counseling about Latent TB, assess risks of reactivation, and considerations for treatment for latent tuberculosis.
- Refer to Infectious Disease for counseling and management.
- Order a two-view chest X-ray to rule out active TB.
- Both C and E.

The answer is f.



Discussion



Neither the QuantiFERON Gold nor the PPD skin test is used to diagnose an active TB infection. This patient has latent TB, and the next steps to embark on are to rule out active TB and assess the need for latent TB infection (LTBI) treatment.

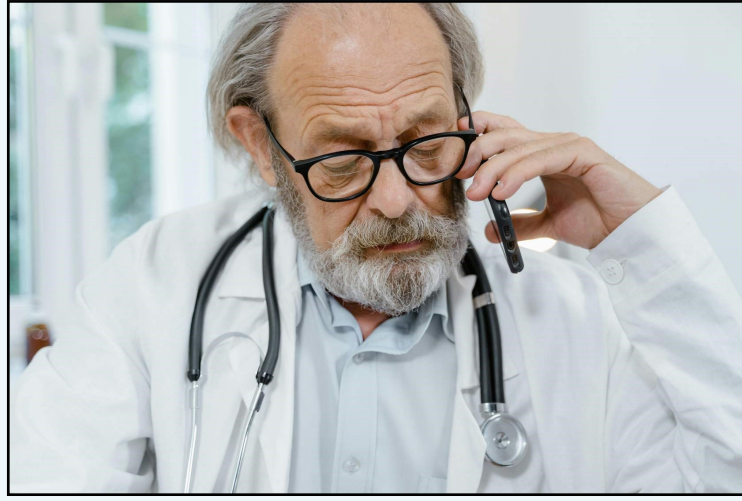
Unless the patient is a transplant patient or is otherwise immunocompromised, patients with latent TB infection should not be routinely referred to infectious disease for management.

This patient is asymptomatic and appears to have been exposed to TB but does not have any symptoms consistent with active TB (see ROS above). If any symptoms of active TB are present, a two-view chest X-ray should be ordered in addition to "AFB sputum," and the patient should be referred to Infectious Disease for treatment.

Repeating TB testing is not indicated or required, and patients with a history of the BCG vaccine should never be given a PPD skin test to screen for TB, as it will always be positive.

Initial Evaluation

- Clinical Assessment and History:** Evaluate for any symptoms suggestive of active TB, such as persistent cough, hemoptysis, night sweats, weight loss, or fever. Assess epidemiologic risk factors, including recent exposure to active TB, travel to TB-endemic areas, and immunosuppressive conditions.
- Chest Radiography:** Perform a **two-view** chest X-ray to exclude active pulmonary TB. If the chest radiograph is abnormal, further diagnostic testing, including sputum AFB and referral to ID, may be warranted. If you encounter a patient with a negative chest X-ray and is asymptomatic, please add LTBI to the problem list.
- HIV Testing:** The American Thoracic Society, Centers for Disease Control and Prevention, and Infectious Diseases Society of America recommend routine HIV testing for all patients with a positive TB test due to the increased risk of progression to active TB in HIV-infected individuals.



Counseling Considerations

- Risk of Progression:** Discuss the risk factors for progression from LTBI to active TB, such as immunosuppression, diabetes, chronic kidney disease, and other comorbid conditions. If there is a question about the need for LTBI treatment, you may send a Dr. Advice to Infectious Disease in the patient's home area to clarify any clinical questions. Diabetics have a 1.6-fold increased risk of progression to active TB compared to the general population. Those with ESRD have a 9.8-fold higher risk.
- Treatment Options:** You can use the **SMARTRX ADULT, Latent Tuberculosis Infection** Smart Set (see snapshot below) for guidance on preferred and alternative medication options, dosing, medications preferred in pregnancy, cautions/precautions with medication treatment, and lab monitoring. Explain the treatment regimens for LTBI, emphasizing the importance of adherence to prevent the development of active TB.

Preferred regimens include 3 months of once-weekly rifapentine plus isoniazid or 4 months of daily rifampin. Advise the patient that there may be medication interactions with these treatments. (You can use the Smart Set to add the medication and use the **INTERACTION CHECKER** to give patient guidance on their particular situation.)

SMARTRX - ADULT, Latent Tuberculosis Infection (LTBI)			
Adult Latent Tuberculosis - Treatment Options			
Preferred	Alternate		
RIF x 4 months (first line) RIFAMPIN (RIF) 10 mg/kg daily (max: 600 mg)	INH + RIF x 3 months ISONIAZID (INH) 5 mg/kg daily (max: 300 mg) + RIFAMPIN (RIF) 10 mg/kg daily (max: 600 mg) + VITAMIN B6 (pyridoxine) 50 mg daily	INH + RPT x 12 weeks (not studied in pregnancy) ISONIAZID (INH) 15 mg/kg weekly (max: 900 mg) + RIFAPENTINE (RPT) 32.1 to 49.9 kg: 750 mg weekly ≥ 50 kg: 900 mg weekly (max: 900 mg) + VITAMIN B6 (pyridoxine) 50 mg daily	INH-6H (6 mos) or INH-9H (9 mos)* ISONIAZID (INH) 5 mg/kg daily (max: 300 mg) + VITAMIN B6 (pyridoxine) 50 mg daily
RIF: - Many patients experience nausea and/or headache with Rifampin, which typically self-resolve. If symptoms persist, patients may take Tylenol for headache and Zofran for nausea - Advise patient that Rifampin turns bodily fluids (i.e., tears, urine) orange, and stains contact lenses - Risk of flu-like syndrome - Advise patients on hormonal birth control that Rifampin can decrease effectiveness; need to use a non-hormonal form of birth control while on Rifampin - If AST/ALT rises to 5x ULN, or 3x ULN with concomitant symptoms (i.e., abdominal pain and/or anorexia/nausea/vomiting, dark urine, easy bleeding/bruising, jaundice), treatment should be stopped immediately. Both ALT/AST needed every 4 weeks	RPT: - RPT is in shortage, order after confirming supply - Dosage based on patient weight - Advise patient that RPT turns bodily fluids (i.e., tears, urine) orange, and stains contact lenses - Risk of flu-like syndrome	- Prescribe 3 months at a time - While 9H was estimated to be more efficacious than 6H at reducing the risk of TB in one analysis, 6H has been studied more extensively, may be more cost-effective than 9H, and may have lower risk of hepatotoxicity. Nonetheless, 9H is probably more efficacious and is commonly recommended for patients who are immunosuppressed, tolerating treatment at 6 months, and willing to continue. [INSTE/NTCA Clinical Recs, p.36]	
INH: - INH is often given if adverse effects to Rifampin occurs such as elevated AST/ALT, or if drug interactions preclude rifampin - Monitor for symptoms of liver toxicity (nausea, abdominal pain, jaundice...). Labs are ordered every 4 weeks while on treatment			
Reference(s): - CDC LTBI 2020 Guidelines - CDC LTBI Treatment Regimens table - Testing and Treatment of LTBI in the US - Clinical Recs (Feb 2021): National Society of Tuberculosis Clinicians/National Tuberculosis Controllers Association (NSTC/NTCA)			
- PNL LTBI RIF (4 MOS), rifAMPin 300 mg cap (RIFADIN), (10 mg/kg: >=60 kg), 2 daily + labs - PNL LTBI INH + RIF (3 MOS), isoniazid 300 mg tab (NYDRAZID), (5 mg/kg: >=60 kg), 1 daily, + rifAMPin 300 mg cap (RIFADIN), (10 mg/kg: >=60 kg), 2 daily + OTC pyridoxine 50 mg tab (VITAMIN B6), 1 daily + labs - PNL LTBI (12 WKS), isoniazid (NYDRAZID), (15 mg/kg), select formulation + rifapentine 150 mg tab (PRIFTIN), 900 mg every week + OTC pyridoxine 50 mg tab (VITAMIN B6), 1 daily + labs - PNL LTBI INH (6 MOS), isoniazid 300 mg tab (NYDRAZID), (5 mg/kg: >=60 kg), 1 daily + OTC pyridoxine 50 mg tab (VITAMIN B6), 1 daily + labs - PNL LTBI INH (9 MOS), isoniazid 300 mg tab (NYDRAZID), (5 mg/kg: >=60 kg), 1 daily + OTC pyridoxine 50 mg tab (VITAMIN B6), 1 daily + labs			

- Monitoring and Follow-Up:** If the patient is interested in treatment, order the recommended HIV, AST, ALT, and CBC with diff and ask patient to follow up with PCP for treatment ordering (do not refer to ID). Because of the episodic nature of our workstream on GCN, it is not recommended to start LTBI treatment virtually, as the patient would be required to have monthly clinical and lab follow-up to monitor for medication-adverse effects and adherence. If pretreatment aminotransferase levels are normal, further laboratory monitoring may be limited to clinical indications.

Visual Context

The following table provides a risk-based approach for LTBI testing and treatment, highlighting the importance of epidemiologic and host risk factors in guiding management decisions.

Table 1. Risk-Based Approach for Latent Tuberculosis Infection (LTBI) Testing and Treatment.*	
Findings Based on History	Recommended Action
Any of the following high-priority risk factors†: Household or other close contact with someone with infectious active tuberculosis disease‡; Birth, residence, or prolonged travel (>1 mo) in a setting in which tuberculosis disease is common (most countries in Africa, Asia, Eastern Europe, Latin America, and the Pacific Islands)§; Other circumstances based on local epidemiology (e.g., corrections facilities and homeless shelters)§; Immunosuppression (HIV infection or current or planned immunosuppression with the use of a TNF-α antagonist, glucocorticoids equivalent to prednisone at a dose of 2 mg per kilogram of body weight per day for ≥1 mo, or other immunosuppressing medications)‡	Testing for <i>M. tuberculosis</i> infection should be performed. If LTBI is present, LTBI treatment is recommended unless there are medical contraindications or clinically significant drug-drug interactions that cannot be mitigated.
Any of the following biologic factors that confer generally modest increased risk of progression to active tuberculosis disease if <i>M. tuberculosis</i> infection is present‡: diabetes mellitus; chronic kidney disease; leukemia or lymphoma; cancer of the head or neck; chronic malabsorption, gastrectomy, or intestinal bypass; a body-mass index (the weight in kilograms divided by the square of the height in meters) of ≤20; silicosis; current or former smoking status; or an age of ≤5 yr¶	In the absence of epidemiologic risk factors (as noted above) for tuberculosis exposure, testing for <i>M. tuberculosis</i> infection is generally not recommended. However, if LTBI is present, LTBI treatment is recommended unless there are medical contraindications or clinically significant drug-drug interactions that cannot be mitigated.
No epidemiologic risk factors for tuberculosis exposure and no host risk factors for progression to active tuberculosis disease if <i>M. tuberculosis</i> infection is present	Testing for <i>M. tuberculosis</i> infection is not recommended.

* This table is based on recommendations from the National Society of Tuberculosis Clinicians.⁶ HIV denotes human immunodeficiency virus, and TNF-α tumor necrosis factor α.

† Tuberculin skin test (TST) results are interpreted at the listed positivity thresholds on the basis of epidemiologic or host risk and to improve test positive and negative predictive values. Interferon-γ release assays currently have absolute thresholds for positivity without incorporation of host or epidemiologic risk.

‡ The TST positivity threshold is 5 mm or more.

§ The TST positivity threshold is 10 mm or more.

¶ Children 5 years of age or younger are at high risk for rapid progression to active tuberculosis disease after infection, and severe forms of tuberculosis (including disseminated and central nervous system tuberculosis) are more likely to develop in such children than in older children or in adults.

Note: From Latent Tuberculosis Infection [Infographic], by Maunank Shah and Susan Dorman, 2021, *The New England Journal of Medicine*. (<https://www.nejm.org/doi/full/10.1056/NEJMcp2108501>).

References

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